

Beyond Homogeneous Adaptation: Decoupled Control for Sample- and Layer-Aware Test-Time Robustness in Temporally Correlated Streams

Haoyu Xiong, Yang Li, Chengchao Wang, Qiuxia Yang, Wenhan Wu, Zhengpeng Zhao, Jinde Cao, *Fellow, IEEE*, and Yuanyuan Pu

Abstract—Test-Time Adaptation (TTA) enables pre-trained models to adapt online to distribution shifts during inference, but its self-training dynamics are fragile under temporally correlated streams. Once biased target evidence is used for adaptation, the resulting errors can accumulate over time, distort target representations, and persist unevenly across network layers. Existing methods typically mitigate this problem with confidence filtering or global parameter regularization, but these homogeneous controls overlook two forms of heterogeneity: target samples differ in whether they provide trustworthy adaptation evidence, and layers differ in whether they can safely absorb the induced updates. We propose DCF, a Decoupled Control Framework that stabilizes TTA by jointly controlling *which samples drive adaptation* and *which layers retain the resulting update*. At the sample level, DCF uses Probe-supported Sample Routing (PSR) to identify probe-supported trusted evidence via entropy and a Fourier stress response, while Routed Geometry Repair (RGR) reuses routed-away samples for optimal-transport-based geometry repair instead of discarding them. At the parameter level, the sample-side update is treated as only a candidate update: Curvature-aware Layer Retention (CLR) then performs layer retention, preserving plasticity in layers and suppressing updates in layers that exhibit high source mismatch. This route-adapt-retain design prevents unreliable evidence from dominating adaptation while avoiding indiscriminate rollback of useful plasticity. Extensive experiments on temporally correlated corruptions, natural shifts, and long-horizon streams demonstrate that DCF consistently improves test-time robustness and prevents the late-stage collapse of existing TTA methods. Code is available at <https://github.com/ioslide/DCF>.

Index Terms—Deep learning, test-time robustness, test-time adaptation, domain shift.

I. INTRODUCTION

Deep neural networks [1, 2] have achieved remarkable performance when training and test data are independently and

This work was supported by the National Natural Science Foundation of China (61271361, 61761046, 62162068, 52102382, and 62362070); the Applied Basic Research Program of Yunnan Province (202001BB050043, 202401AS070149); the Major Science and Technology Special Project of Yunnan Province (202302AF080006); the Yunnan University School of ISE Graduate Student Scientific Research Project (ISEGS2026A03, ISEGS2026B03); the Yunnan Provincial Department of Education Science Research Fund (2026Y0180); and the Yunnan Key Laboratory of Low-light Night Vision Technology and Intelligent Visual Navigation (202449CE340004).

H. Xiong, Y. Li, C. Wang, and Z. Zhao are with the School of Information Science and Engineering, Yunnan University, Kunming 650500, China. Q. Yang, W. Wu, and Y. Pu are with the School of Engineering, Yunnan University, Kunming 650500, China. Y. Pu is also with the Key Lab. of IoT Technology and Application in Yunnan Province, China. J. Cao is with Southeast University, Nanjing 210096, China, and Yonsei University, Seoul 03722, South Korea. The first two authors contributed comparably to this work. Corresponding authors: Z. Zhao and Y. Pu (e-mail: zhpzhao@ynu.edu.cn; yuanyuanpu@ynu.edu.cn).

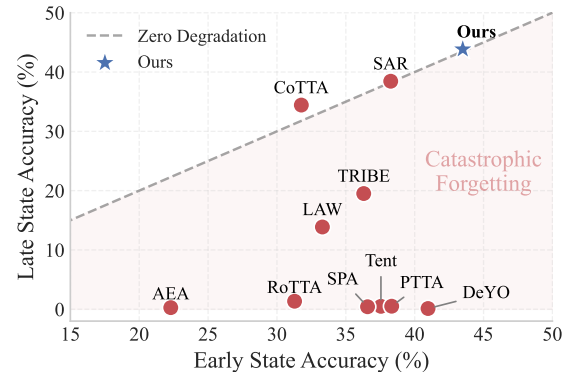


Fig. 1. **Long-horizon instability of existing TTA methods.** Each point represents a method's average accuracy at the early stage (x-axis, Domain index 15) and late stage (y-axis, Domain index 225) of adaptation on a temporally correlated stream. The diagonal line indicates equal performance at both stages. Many methods that achieve strong early performance later fall below the diagonal, indicating unstable long-horizon adaptation.

identically distributed (i.i.d.). However, their reliability often degrades sharply once deployment environments deviate from the training distribution [3–7]. In practical visual recognition systems, test samples rarely arrive as randomly shuffled instances from a fixed target domain. Instead, they are often observed as temporally correlated streams, where sensor noise, illumination, weather, corruption type, rendering style, and scene statistics evolve gradually over time [8]. Test-time adaptation (TTA) [9–12] addresses this deployment setting by updating a pre-trained source model online using only unlabeled test samples, making it an appealing paradigm for source-free robustness under distribution shift.

Despite its promise, online adaptation introduces a critical instability that is especially severe in temporally correlated streams. Since the model adapts from its own predictions, local mistakes may be repeatedly reinforced by subsequent updates. As adaptation continues, biased target evidence can distort feature representations, alter the model's sensitivity to future samples, and eventually accumulate into long-horizon performance degradation. As shown in Figure 1, several TTA methods achieve competitive accuracy in the early stage of adaptation but deteriorate substantially in the late stage. This phenomenon raises a central question:

Why does test-time adaptation under temporally correlated shifts become unstable over time?

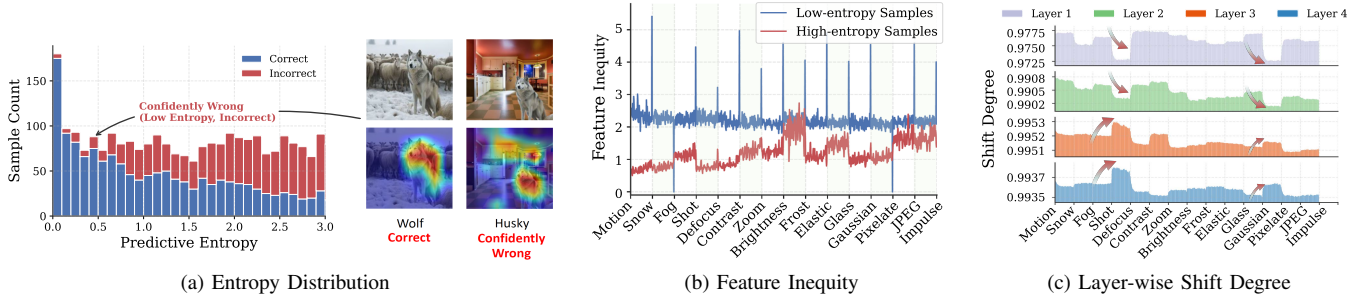


Fig. 2. **Why homogeneous TTA fails.** (a) Low entropy does not guarantee reliability: even highly confident predictions can be wrong and may be driven by spurious cues. (b) High-entropy samples are weak pseudo-label candidates, but their *centroid entropy deficit* indicates that they still capture shifting target geometry. Discarding them accelerates representation bias and removes useful structural anchors. (c) Layer drift, measured by KL divergence from the source model, is heterogeneous across depth, with different magnitudes and even opposite trends.

Existing methods [9, 13–21] typically address this instability through two forms of homogeneous control. The first filters samples according to prediction confidence, implicitly assuming that confident predictions provide reliable pseudo-label supervision. The second regularizes parameter movement, implicitly assuming that adaptation-induced drift can be controlled by a global or nearly uniform stabilization rule. Although these strategies are effective in certain settings, they do not fully explain or resolve the failure pattern observed in long temporally correlated streams.

We argue that the instability of temporally correlated TTA is not merely caused by pseudo-label noise or parameter drift in isolation. Instead, it is better understood as a coupled sample–layer feedback process. At the sample level, unreliable target evidence may induce biased self-training updates. At the parameter level, these biased updates are not absorbed uniformly across network depth: some layers may require plasticity to adapt to benign appearance changes, whereas others may suffer from semantic forgetting when pushed away from the source model. Once retained in source-sensitive layers, such drift changes the model’s future predictions and makes subsequent sample selection even more biased. This feedback loop can gradually amplify local adaptation errors into long-horizon collapse.

This coupled failure process reveals two overlooked forms of heterogeneity in online TTA. First, target samples differ in whether they provide trustworthy adaptation evidence. A low-entropy prediction is not necessarily reliable: it may be driven by shortcut-sensitive cues and still produce a systematically biased update. Conversely, high-entropy samples may be unsafe for direct pseudo-label supervision but can still encode useful structural information about the evolving target distribution. Second, network layers differ in whether they can safely retain the induced update. Some layers should remain plastic, while others should be constrained toward the source model to prevent irreversible semantic drift. Therefore, stable TTA requires joint control over *which samples drive adaptation*, *how non-trusted samples are reused*, and *which layers retain the resulting update*.

We summarize these findings into three observations:

- **O1: Confidence is not reliability.** Low-entropy predictions can still be incorrect when they are supported by spurious or

shortcut-sensitive cues. Thus, entropy alone is an insufficient criterion for selecting safe adaptation evidence (Figure 2a).

- **O2: Uncertainty is not useless.** Although high-entropy samples are unreliable for hard pseudo-label supervision, they may still contain target-domain structural information. Tracking their *centroid entropy deficit* shows that routed-away samples can reflect the evolving target geometry. Discarding them may accelerate representation bias and remove useful structural anchors (Figure 2b).
- **O3: Drift is not uniform across depth.** Adaptation-induced drift is strongly layer-dependent, with different magnitudes and even opposite trends across layers. Therefore, a single global stabilization rule is too coarse to prevent long-horizon degradation (Figure 2c).

Motivated by these observations, we propose DCF, a Decoupled Control Framework for stable test-time adaptation in temporally correlated image streams. The key idea is to decouple online adaptation into three control stages: *route*, *adapt*, and *retain*. This design separates three decisions that are often entangled in previous TTA methods: which samples should provide label-like adaptation evidence, how routed-away samples should be reused without hard pseudo-labeling, and where the resulting parameter update should be allowed to persist. By making these decisions separately, DCF prevents unreliable evidence from dominating adaptation while avoiding indiscriminate rollback of useful plasticity.

Concretely, DCF follows a route–adapt–retain principle. First, Probe-supported Sample Routing (PSR) routes target samples into trusted and routed-away subsets by combining predictive entropy with a Fourier stress response. This stress response probes whether a confident prediction is supported by adaptation-relevant evidence rather than by stress-inert shortcut cues. Second, Routed Geometry Repair (RGR) performs decoupled sample-side adaptation: trusted samples provide conservative consistency signals, whereas routed-away samples are reused to repair target feature geometry through soft optimal-transport-based alignment. Third, Curvature-aware Layer Retention (CLR) treats the sample-side update as only a candidate update and selectively retains it at the layer level. Layers whose curvature profiles remain compatible with the source model are allowed to preserve plasticity, while layers with high source mismatch are pulled back toward the source

anchor. Through this route–adapt–retain design, DCF mitigates the coupled sample–layer feedback that causes long-horizon TTA instability.

Our contributions are summarized as follows:

- We provide a systematic analysis of long-horizon failure in temporally correlated TTA and show that instability arises from a coupled sample–layer feedback process rather than from pseudo-label noise or parameter drift alone.
- We reveal three forms of overlooked heterogeneity in online TTA: confident predictions can be unreliable, routed-away uncertain samples can still preserve target geometry, and adaptation-induced drift is highly layer-dependent.
- We introduce DCF, a Decoupled Control Framework that implements a route–adapt–retain design by decoupling trusted-evidence routing, routed-away geometry repair, and curvature-aware layer retention.
- We conduct extensive experiments on synthetic corruptions, natural distribution shifts, label-shifted streams, mixed-severity streams, and long-horizon temporally correlated streams, demonstrating that DCF consistently improves robustness and prevents the late-stage collapse observed in strong TTA baselines.

II. RELATED WORK

We relate DCF to existing adaptation methods without and with target data, and further discuss sample reliability estimation, target-geometry preservation, and parameter stabilization for online test-time adaptation.

Adaptation without Target Data. Mitigating distribution shifts has been widely studied at training time, including domain generalization (DG) [22], robust representation learning [23], and data augmentation strategies [24]. These methods aim to improve out-of-distribution robustness by learning source models that generalize to unseen test environments. However, deployment shifts are difficult to enumerate in advance, and training-time strategies usually require additional source-side objectives, augmentations, or training costs. In contrast, test-time adaptation updates the model directly from unlabeled target samples during deployment, making it attractive for source-free robustness under evolving test streams.

Adaptation with Target Data. Methods that exploit target data include offline unsupervised domain adaptation and online test-time adaptation.

• **Unsupervised domain adaptation (UDA).** Conventional UDA assumes access to labeled source data and unlabeled target data during adaptation, and reduces domain discrepancy through feature or moment alignment [25–28], adversarial learning [29, 30], pseudo-label/self-training strategies [31–33], or optimal transport [4]. Source-free UDA further removes source data during adaptation and often relies on information maximization [5], pseudo-label refinement [34], or classifier-discrepancy/self-training objectives [32, 35]. Although effective when the target domain is available in advance, these methods usually adapt offline over the whole target set and often require multiple training epochs. Therefore, they are less suitable for online test streams, where samples arrive sequentially and the target distribution may evolve over time.

• **Test-time adaptation (TTA).** TTA adapts a pre-trained model during inference using only unlabeled test samples. Some methods modify source training by introducing auxiliary objectives, as in test-time training [10]. Fully test-time adaptation does not alter source training and can be directly applied to a pre-trained model. Representative methods update batch-normalization statistics [36], minimize prediction entropy [9, 13], enforce consistency regularization [17], or use memory and teacher models for online adaptation [17, 18]. Recent studies further show that online TTA can be unstable under realistic streams, especially with temporal correlation, small batches, mixed domains, or imbalanced label distributions. NOTE [37] highlights the effect of temporal correlation, while continual and practical TTA methods improve robustness through teacher models, memory banks, reset strategies, normalization correction, or prior correction [15–18, 20, 38, 39]. These methods significantly improve online robustness, but most of them still stabilize adaptation through relatively homogeneous sample filtering or global update control. In this paper, we mainly focus on fully test-time adaptation and study its long-horizon instability from a coupled sample–layer perspective.

Reliability- and Stability-aware TTA. A central challenge in online TTA is to decide which unlabeled samples should drive adaptation and how the induced parameter updates should be stabilized over time. Existing methods often address the sample side by estimating reliability from confidence or entropy, assuming that low-entropy predictions provide trustworthy pseudo-label supervision. For example, EATA [13] and SAR [14] filter redundant or unreliable samples through entropy-based criteria, DeYO [40] introduces shape-aware sample selection beyond entropy, and ETAGE [21] further incorporates gradient-norm information to detect unstable samples. However, confidence does not necessarily imply reliability under temporally correlated shifts: a confident prediction may still be supported by shortcut-sensitive cues, and repeatedly adapting to such samples can reinforce confirmation bias. Another line of work stabilizes TTA from the parameter side. CoTTA [17] uses stochastic restoration to mitigate catastrophic drift, RoTTA [18] combines robust memory sampling with teacher-student updates, and LAW [41] studies learning-rate-based mechanisms for sustained adaptation. These methods reduce long-term drift, but their stabilization rules are usually global or nearly uniform across the network, overlooking that different layers may respond to distribution shifts in substantially different ways [42].

Overall, existing TTA methods mainly stabilize online adaptation through sample filtering or global update control. In contrast, DCF attributes long-horizon instability to coupled sample–layer feedback, and decouples sample routing, routed-away geometry repair, and layer-wise update retention.

III. METHODOLOGY

A. Problem Setup

Test-time adaptation starts from a source model $f_{\theta_0} : \mathcal{X} \rightarrow \Delta^{K-1}$ trained on a labeled source distribution $\mathcal{P}_0(X, Y)$. At deployment, the model receives an unlabeled test stream $\{x_t\}_{t=1}^T$, where $x_t = \{x_{t,i}\}_{i=1}^{N_t}$ denotes the mini-batch of

size N_t observed at time t and $x_{t,i} \sim \mathcal{P}_t(X)$. The target distribution $\mathcal{P}_t$ may evolve over time because of temporally correlated covariate shifts, such as changing corruption types, sensor noise, illumination, weather, or rendering style. It may also involve label-distribution changes, denoted by the class prior $\pi_t^Y(\cdot) := \mathcal{P}_t(Y = \cdot)$. We write $\theta_t = \{\theta_t^l\}_{l=1}^L$ for the layer-wise parameters of the online model at time t .

The goal of TTA is to update the model online using only the current unlabeled mini-batch before producing final predictions on that batch. A common adaptation step can be written as

$$\theta_{t+1}^l := \theta_t^l - \eta \nabla_{\theta_t^l} \mathcal{L}_{\text{TTA}}(x_t; \theta_t), \quad (1)$$

where $\mathcal{L}_{\text{TTA}}$ is usually derived from the model's own predictions, such as entropy minimization or consistency regularization. Under temporally correlated streams, however, such an update can become unstable: unreliable pseudo-labels may induce biased gradients, and the resulting parameter movement may persist in layers that are sensitive to source-domain forgetting. Therefore, stable TTA requires not only deciding *which samples* should generate adaptation signals, but also deciding *where* the induced update should be retained.

B. Decoupled Control View of DCF

We formulate DCF as a decoupled online control rule for temporally correlated TTA. At each test-time step, the model must make two coupled decisions: *which target samples should be trusted to drive adaptation* and *which layers should be allowed to retain the resulting parameter update*. Conventional TTA methods often merge these decisions into one homogeneous update: selected samples optimize a single objective, and the induced update is directly applied to the adapted parameters. This coupling is unsafe under temporally correlated shifts because unreliable samples can bias the update, while even a useful sample-side update may be harmful if it persists uniformly in source-sensitive layers.

To decouple these decisions, DCF follows a route-adapt-retain principle. First, it *routes* target samples into a trusted set and a routed-away set. Second, it *repairs* target geometry by using routed-away samples as structural information rather than discarding them. Third, it *retains* only the layer-wise parts of the candidate update that remain compatible with the source model.

Given the current mini-batch x_t , DCF constructs a sample-side candidate update by minimizing two complementary losses:

$$\theta_t^+ = \theta_t - \eta \nabla_{\theta_t} (\mathcal{L}_{\text{trust}}(\mathcal{R}_t) + \mathcal{L}_{\text{geo}}(\mathcal{U}_t)), \quad (2)$$

where $\mathcal{R}_t$ denotes the trusted set and $\mathcal{U}_t$ denotes the routed-away set. The trusted loss $\mathcal{L}_{\text{trust}}$ is induced by PCS-weighted consistency learning on probe-supported samples, whereas the geometry loss $\mathcal{L}_{\text{geo}}$ is induced by Routed Geometry Repair (RGR) on routed-away samples.

Importantly, θ_t^+ is not directly accepted as the final adapted model. It is treated as a candidate update whose persistence must be controlled layer by layer:

$$\theta_{t+1} = \text{Retain}_{\mu_t, I_t, I_0}(\theta_t^+, \theta_0), \quad (3)$$

where $\text{Retain}_{\mu_t, I_t, I_0}$ is implemented by Curvature-aware Layer Retention (CLR). This final retention step prevents a globally useful sample-side update from being indiscriminately preserved in layers that exhibit high source mismatch.

C. Route: Probe-supported Sample Routing

The first control decision is sample routing: how the current target batch should be partitioned before any adaptation objective is applied. Prediction confidence alone is insufficient for this decision. A low-entropy prediction may still be supported by shortcut-sensitive features and can therefore reinforce confirmation bias under repeated updates. We address this by combining entropy with a targeted Fourier stress probe. The probe creates a counterfactual stressed view that preserves the semantic layout while perturbing frequency-direction components. A sample is routed to the trusted set only when it is both confident and sufficiently responsive to this structured stress, indicating that its prediction is supported by adaptation-relevant evidence rather than stress-inert shortcuts.

Fourier-basis stress probe. At test-time step t , we apply a small additive Fourier-basis perturbation to the current batch x_t to obtain a stressed batch $x_t^s = \mathcal{A}^s(x_t)$:

$$\begin{aligned} \Phi_{f,\omega}(i, j) &= R \sin\left(2\pi f(i \cos \omega + j \sin \omega - \pi/4)\right), \\ [\mathcal{A}^s(x_t)]_{i,j,c} &= \text{Clamp}_{[0,1]} \left([x_t]_{i,j,c} + \sigma_c \Phi_{f_c, \omega_c}(i, j) \right), \quad (4) \\ c &\in \{1, 2, 3\}. \end{aligned}$$

Here, R is chosen such that $\|\Phi_{f,\omega}\|_2 = 1$. For each channel c , we independently sample the frequency and direction as $f_c \sim \mathcal{U}[1, 224]$ and $\omega_c \sim \mathcal{U}[0, \pi]$, and sample the perturbation strength as $\sigma_c \sim \text{Exp}(1/\lambda)$.

Given a sample $x_{t,i} \in x_t$ and its stressed counterpart $x_{t,i}^s = \mathcal{A}^s(x_{t,i})$, we define the predicted class on the clean view and the *Perturbation Consistency Score* (PCS) as

$$\begin{aligned} x_{t,i}^s &= \mathcal{A}^s(x_{t,i}), \quad \hat{y}_t(x_{t,i}) = \arg \max_y p_{\theta_t}(y | x_{t,i}), \\ s_{\theta_t}(x_{t,i}) &= [p_{\theta_t}(\hat{y}_t(x_{t,i}) | x_{t,i}) - p_{\theta_t}(\hat{y}_t(x_{t,i}) | x_{t,i}^s)]_+. \quad (5) \end{aligned}$$

Here $[a]_+ := \max(a, 0)$. PCS measures how strongly the predicted-class confidence responds to controlled Fourier stress. A high PCS indicates that the prediction is stress-responsive and provides stronger evidence for adaptation, whereas a low PCS suggests that a confident prediction may be supported by stress-inert shortcuts.

Sample routing. Let $E_{\theta_t}(x_{t,i})$ denote the predictive entropy of sample $x_{t,i}$. We define the trusted set as

$$\mathcal{R}_{\theta_t}(x_t) = \left\{ x_{t,i} \in x_t \mid \begin{array}{l} E_{\theta_t}(x_{t,i}) < v_{\text{Ent}}, \\ s_{\theta_t}(x_{t,i}) > v_{\text{PCS}} \end{array} \right\}, \quad (6)$$

where v_{Ent} and v_{PCS} are thresholds for confidence and stress response. The routed-away set is

$$\mathcal{U}_{\theta_t}(x_t) = x_t \setminus \mathcal{R}_{\theta_t}(x_t). \quad (7)$$

For brevity, we write $\mathcal{R}_t = \mathcal{R}_{\theta_t}(x_t)$ and $\mathcal{U}_t = \mathcal{U}_{\theta_t}(x_t)$ in the following sections. The next section specifies how these two routed subsets are used with different adaptation objectives.

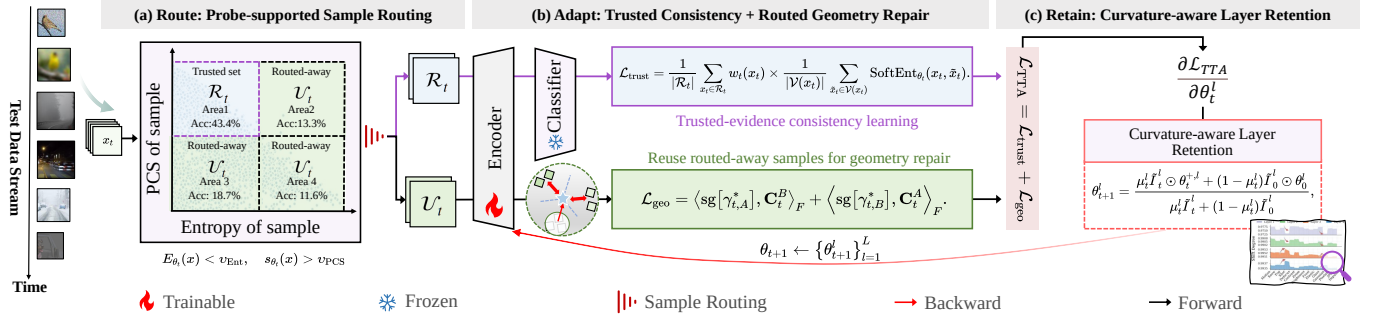


Fig. 3. **Overview of DCF.** DCF implements a route-adapt-retain pipeline. Probe-supported Sample Routing routes target samples into trusted and routed-away subsets using entropy and a Fourier stress response. Trusted samples drive PCS-weighted consistency learning, whereas routed-away samples are reused by Routed Geometry Repair (RGR) for geometry repair. The resulting candidate update is then selectively retained by Curvature-aware Layer Retention (CLR) through curvature-aware layer-wise retention.

D. Adapt: Decoupled Sample-side Adaptation

After sample routing, DCF performs decoupled sample-side adaptation. The key point is that $\mathcal{R}_t$ and $\mathcal{U}_t$ should not be optimized with the same supervision signal. Trusted samples are allowed to provide label-like consistency supervision because they are both low-entropy and stress-responsive. Routed-away samples are not discarded, but they are also not converted into hard pseudo-labels; instead, they are reused to repair target feature geometry. The resulting sample-side objective is $\mathcal{L}_{\text{trust}} + \mathcal{L}_{\text{geo}}$, which produces the candidate update in Eqn. (2).

Trusted-evidence consistency learning. For each trusted sample $x_{t,i} \in \mathcal{R}_t$, we use three views: the clean view $x_{t,i}$, a weak semantics-preserving augmentation $x_{t,i}^w = \mathcal{A}^w(x_{t,i})$, and the Fourier-stressed view $x_{t,i}^s = \mathcal{A}^s(x_{t,i})$. Let

$$\mathcal{V}(x_{t,i}) = \{x_{t,i}, x_{t,i}^w, x_{t,i}^s\}.$$

Each augmented view is trained to remain consistent with the original one:

$$\text{SoftEnt}_{\theta_t}(x_{t,i}, \tilde{x}_{t,i}) := - \sum_{k=1}^K p_{\theta_t}(k | x_{t,i}) \log p_{\theta_t}(k | \tilde{x}_{t,i}), \quad (8)$$

where $\tilde{x}_{t,i} \in \mathcal{V}(x_{t,i})$.

To make PCS weighting well-defined for small or degenerate trusted sets, we first compute the minimum and maximum PCS values within $\mathcal{R}_t$:

$$s_{\min} = \min_{x_{t,i} \in \mathcal{R}_t} s_{\theta_t}(x_{t,i}), \quad s_{\max} = \max_{x_{t,i} \in \mathcal{R}_t} s_{\theta_t}(x_{t,i}).$$

We then define the normalized PCS weight as

$$w_t(x_{t,i}) = \frac{|\mathcal{R}_t| \exp\left(\frac{\tau[s_{\theta_t}(x_{t,i}) - s_{\min}]}{s_{\max} - s_{\min} + \varepsilon_w}\right)}{\sum_{x_{t,j} \in \mathcal{R}_t} \exp\left(\frac{\tau[s_{\theta_t}(x_{t,j}) - s_{\min}]}{s_{\max} - s_{\min} + \varepsilon_w}\right)}. \quad (9)$$

Here τ controls the sharpness of PCS weighting, and $\varepsilon_w = 10^{-6}$ prevents division by zero. The normalization keeps the average weight close to one, so PCS changes relative contributions without changing the overall loss scale.

The trusted-evidence consistency loss is defined as the PCS-

weighted consistency loss over the trusted set:

$$\mathcal{L}_{\text{trust}} = \frac{1}{|\mathcal{R}_t|} \sum_{x_t \in \mathcal{R}_t} w_t(x_t) \times \frac{1}{|\mathcal{V}(x_t)|} \sum_{\tilde{x}_t \in \mathcal{V}(x_t)} \text{SoftEnt}_{\theta_t}(x_t, \tilde{x}_t). \quad (10)$$

When $|\mathcal{R}_t| = 0$, we set $\mathcal{L}_{\text{trust}} = 0$.

Routed-away geometry repair. The trusted objective prevents unreliable pseudo-labels from dominating adaptation, but routing also creates a second challenge: the routed-away samples still describe the evolving target distribution. Simply discarding them removes structural information and biases adaptation toward a small subset of confident samples. Therefore, DCF does not use $\mathcal{U}_t$ for label-like supervision. Instead, Routed Geometry Repair (RGR) reuses these samples for geometry repair.

The goal of RGR is to preserve target structural diversity without converting unreliable samples into hard pseudo-labels. We formulate this as an optimal-transport alignment problem between routed-away features and source-initialized class centroids. The transport constraints discourage assignments from collapsing to a few dominant classes, while source-initialized centroids provide stable anchors without requiring access to source training samples. We detail this repair procedure through the following internal steps.

Feature concentration diagnostic. We quantify representation bias by the *centroid entropy deficit* $I(\mathbf{Z})$. For a batch of normalized features $\mathbf{Z} = \{\mathbf{z}_i\}_{i=1}^B$ with centroid $\bar{\mathbf{z}} = \frac{1}{B} \sum_{i=1}^B \mathbf{z}_i$, this diagnostic measures the entropy deficit of the centroid prediction:

$$I(\mathbf{Z}) = \log K + \sum_{k=1}^K \bar{p}_k \log \bar{p}_k, \quad (11)$$

where $\bar{p} = \text{softmax}(c_{\theta_t}(\bar{\mathbf{z}}))$, and c_{θ_t} denotes the classifier head applied to the feature centroid. A larger $I(\mathbf{Z})$ indicates that the centroid prediction is more concentrated on a small number of classes. This motivates using routed-away samples to repair target geometry rather than removing them from adaptation.

Source-initialized centroids and routed-away features. Let

$n_t := |\mathcal{U}_t|$. We re-index the routed-away samples as

$$\mathcal{U}_t = \{u_{t,i}\}_{i=1}^{n_t}.$$

When $n_t = 0$, we set $\mathcal{L}_{\text{geo}} = 0$ and skip the OT computation. Otherwise, for each routed-away sample, we construct two independent weak stochastic views:

$$u_{t,i}^A = \mathcal{A}^{w_1}(u_{t,i}), \quad u_{t,i}^B = \mathcal{A}^{w_2}(u_{t,i}),$$

where $\mathcal{A}^{w_1}$ and $\mathcal{A}^{w_2}$ are semantics-preserving augmentations. Let

$$\mathbf{z}_{t,i}^V = \frac{h_{\theta_t}(u_{t,i}^V)}{\|h_{\theta_t}(u_{t,i}^V)\|_2}, \quad V \in \{A, B\},$$

denote normalized features.

We maintain shared class centroids $\mathbf{M}_t \in \mathbb{R}^{K \times d}$, initialized from the normalized source classifier weights:

$$\mathbf{M}_{0,c} = \frac{W_c}{\|W_c\|_2},$$

where W_c is the classifier weight of class c . This provides a source-free initialization of class prototypes at test time.

Dynamic-marginal optimal transport. For each view, we compute an entropic OT assignment from routed-away features to the shared centroids. The cosine cost is

$$[\mathbf{C}_t^V]_{ic} = 1 - \cos(\mathbf{z}_{t,i}^V, \mathbf{M}_{t,c}), \quad V \in \{A, B\}. \quad (12)$$

Let $a_t = \mathbf{1}_{n_t}/n_t \in \Delta^{n_t-1}$ be the sample marginal. To make the centroid marginal compatible with possible label shift, we estimate the current target class prior from stop-gradient predictions:

$$\hat{\pi}_t = \frac{1}{N_t} \sum_{i=1}^{N_t} \text{sg}[p_{\theta_t}(\cdot | x_{t,i})], \quad (13)$$

where $\text{sg}[\cdot]$ denotes stop-gradient. We maintain an exponential moving average of this prior:

$$\bar{b}_t = (1 - \rho)\bar{b}_{t-1} + \rho\hat{\pi}_t, \quad \bar{b}_0 = \frac{\mathbf{1}_K}{K}. \quad (14)$$

The final centroid marginal used in the OT constraints is

$$b_t = \rho\bar{b}_t + (1 - \rho)\frac{\mathbf{1}_K}{K}. \quad (15)$$

The uniform component serves as an anti-collapse floor, discouraging routed-away samples from being concentrated on only a few categories.

The transport plan for each view is obtained by

$$\gamma_{t,V}^* = \arg \min_{\gamma \in \Pi(a_t, b_t)} \left\langle \gamma, \mathbf{C}_t^V \right\rangle_F + \varepsilon_{\text{OT}} \sum_{i,c} \gamma_{ic} (\log \gamma_{ic} - 1), \quad V \in \{A, B\}, \quad (16)$$

where

$$\Pi(a_t, b_t) = \{\gamma \in \mathbb{R}_+^{n_t \times K} \mid \gamma \mathbf{1}_K = a_t, \gamma^\top \mathbf{1}_{n_t} = b_t\},$$

and $\varepsilon_{\text{OT}} > 0$ is the entropic regularization strength. The resulting transport plans are detached in the alignment loss; gradients flow through the cross-view costs, but not through the Sinkhorn iterations, the transport constraints, or the dynamic

marginal b_t .

Cross-view geometry repair. Rather than supervising each view by its own assignment or converting routed-away samples into hard pseudo-labels, we enforce cross-view consistency over the soft assignment distribution. The geometry-repair loss is

$$\mathcal{L}_{\text{geo}} = \left\langle \text{sg}[\gamma_{t,A}^*], \mathbf{C}_t^B \right\rangle_F + \left\langle \text{sg}[\gamma_{t,B}^*], \mathbf{C}_t^A \right\rangle_F. \quad (17)$$

This objective encourages two stochastic views of routed-away samples to agree on their structural relationship to the source-initialized centroids. By matching soft assignment geometry rather than hard pseudo-labels, RGR helps maintain structural diversity in the target representation.

Centroid update. The centroids are updated outside backpropagation using the OT-weighted feature mean. First define the accumulated assignment mass

$$m_{t,c} = \sum_{V \in \{A, B\}} \sum_{i=1}^{n_t} \text{sg}[\gamma_{t,V}^*(i, c)].$$

When this mass is non-negligible, the centroid feature mean is

$$\bar{\mathbf{z}}_{t,c} = \frac{1}{m_{t,c}} \sum_{V \in \{A, B\}} \sum_{i=1}^{n_t} \text{sg}[\gamma_{t,V}^*(i, c)] \mathbf{z}_{t,i}^V. \quad (18)$$

Then

$$\mathbf{M}_{t+1,c} \leftarrow \text{norm}((1 - \rho)\mathbf{M}_{t,c} + \rho\bar{\mathbf{z}}_{t,c}). \quad (19)$$

Here $\text{norm}(v) = v/\|v\|_2$, which keeps centroid directions compatible with the cosine cost. If $m_{t,c}$ is too small, we set $\bar{\mathbf{z}}_{t,c} = \mathbf{M}_{t,c}$, so the centroid remains unchanged up to normalization.

E. Retain: Curvature-aware Layer Retention

The decoupled sample-side adaptation step produces a candidate update θ_t^+ , but this update should not be accepted uniformly across all layers. Under temporally correlated shifts, different layers can have different source sensitivity: some layers may need plasticity to absorb low-level appearance changes, whereas deeper semantic layers may suffer from forgetting if the same update is retained without constraint. Thus, the final control decision is not whether to adapt, but where the candidate update should persist.

To implement this decision, Curvature-aware Layer Retention (CLR) performs curvature-aware layer retention. For each layer, it balances two objectives: preserving the candidate update when the layer remains source-compatible, and recovering toward the source anchor when the current curvature profile indicates high mismatch. This yields a layer-wise proximal problem whose closed-form solution is a curvature-weighted barycenter between the candidate parameters and the source parameters.

Curvature-proximal formulation. For each layer l , we choose θ_{t+1}^l by solving

$$\min_{\{\theta_{t+1}^l\}_{l=1}^L} \sum_{l=1}^L \left(\underbrace{\frac{\mu_t^l}{2} \|\theta_{t+1}^l - \theta_t^{+,l}\|_{\tilde{I}_t^l}^2}_{\text{preserve plasticity}} + \underbrace{\frac{1 - \mu_t^l}{2} \|\theta_{t+1}^l - \theta_0^l\|_{\tilde{I}_0^l}^2}_{\text{prevent forgetting}} \right), \quad (20)$$

where $\|v\|_I^2 = v^\top I v$. Here I_t^l and I_0^l denote diagonal curvature proxies for the candidate model and source model, respectively, and

$$\tilde{I}_t^l := I_t^l + \varepsilon_{\text{ret}} \mathbf{1}, \quad \tilde{I}_0^l := I_0^l + \varepsilon_{\text{ret}} \mathbf{1},$$

with $\varepsilon_{\text{ret}} = 10^{-6}$. The layer-wise gate $\mu_t^l \in (0, 1]$ controls how much of the candidate update is retained. A larger μ_t^l keeps the layer closer to $\theta_t^{+,l}$, while a smaller μ_t^l pulls it closer to the source anchor θ_0^l .

Closed-form retention update. Since Eqn. (20) is separable across layers and coordinates, the solution is

$$\theta_{t+1}^l = \frac{\mu_t^l \tilde{I}_t^l \odot \theta_t^{+,l} + (1 - \mu_t^l) \tilde{I}_0^l \odot \theta_0^l}{\mu_t^l \tilde{I}_t^l + (1 - \mu_t^l) \tilde{I}_0^l}, \quad (21)$$

where $\odot$ and division are applied element-wise. This update preserves the candidate parameters when the layer is source-compatible and restores the source anchor when the layer exhibits strong mismatch.

Curvature proxy and retention gate. The raw curvature proxy for the candidate model is estimated from pseudo-label gradients on the current batch:

$$I_t^l = \frac{1}{N_t} \sum_{i=1}^{N_t} g_t^l(x_{t,i}) \odot g_t^l(x_{t,i}), \quad (22)$$

where $g_t^l(x_{t,i}) = \nabla_{\theta_t^{+,l}} \log p_{\theta_t^+}(\hat{y}_t^+(x_{t,i}) \mid x_{t,i})$, $\hat{y}_t^+(x_{t,i}) = \arg \max_y p_{\theta_t^+}(y \mid x_{t,i})$. The source curvature proxy I_0^l is computed analogously at the source parameters θ_0 on the same unlabeled batch, which preserves the source-free test-time setting.

The layer-wise retention gate is

$$\mu_t^l = \mu_0 \exp\left(-\frac{\|\tilde{I}_t^l - \tilde{I}_0^l\|_2^2}{d_l}\right), \quad (23)$$

where $d_l = \dim(\theta^l)$ and μ_0 is the maximum retention strength. When the candidate and source curvature profiles are similar, μ_t^l is large and the candidate update is retained. When the mismatch is large, μ_t^l decreases and the layer is pulled back toward the source model. In this way, CLR turns the sample-side adaptation step into a layer-aware update: useful plasticity is preserved where it is safe, while source-sensitive drift is suppressed before it can accumulate over time.

IV. EXPERIMENTS

A. Setup

Benchmarks. We evaluate DCF on five widely adopted test-time robustness benchmarks that cover both synthetic corruptions and natural distribution shifts. For synthetic shifts, we use ImageNet-C (IN-C) [43], ImageNet-3DCC (IN-3DCC) [44], and ImageNet-C-Bar (IN-C-Bar) [45]. These benchmarks contain diverse corruption patterns and therefore allow us to examine robustness under controlled covariate shifts. For natural shifts, we use ImageNet-R (IN-R) [46] and ImageNet-Sketch (IN-Sketch) [47], which contain images with naturally occurring variations in rendering style, texture, and visual abstraction. Together, these benchmarks provide a comprehensive evaluation

Algorithm 1: Proposed DCF algorithm

Require: Source parameters θ_0 , learning rate η , thresholds $v_{\text{Ent}}, v_{\text{PCS}}$

Input: Unlabeled test stream $\{x_t\}_{t=1}^T$

```

1 Initialize class centroids  $\mathbf{M}_0$  from source classifier;
2 for each mini-batch  $x_t$  at time  $t = 1, 2, \dots, T$  do
    // Stage 1: Route target samples
3   Initialize  $\mathcal{R}_t \leftarrow \emptyset, \mathcal{U}_t \leftarrow \emptyset$ ;
4   for each sample  $x_{t,i} \in x_t$  do
5     Compute predictive entropy  $E_{\theta_t}(x_{t,i})$ ;
6     Compute PCS score  $s_{\theta_t}(x_{t,i})$  via Eqn. (5);
7     if  $E_{\theta_t}(x_{t,i}) < v_{\text{Ent}}$  and  $s_{\theta_t}(x_{t,i}) > v_{\text{PCS}}$  then
8        $\mathcal{R}_t \leftarrow \mathcal{R}_t \cup \{x_{t,i}\}$ ;
9     else
10       $\mathcal{U}_t \leftarrow \mathcal{U}_t \cup \{x_{t,i}\}$ ;
    // Stage 2: Adapt routed subsets separately
11   Initialize  $\mathcal{L}_{\text{trust}} \leftarrow 0, \mathcal{L}_{\text{geo}} \leftarrow 0$ ;
12   if  $|\mathcal{R}_t| > 0$  then
13     Compute  $\mathcal{L}_{\text{trust}}$  on  $\mathcal{R}_t$  via Eqn. (10);
14   if  $|\mathcal{U}_t| > 0$  then
15     Compute dynamic marginal  $b_t$  via Eqn. (15);
16     Solve OT plans  $\gamma_{t,A}^*, \gamma_{t,B}^*$  via Eqn. (16);
17     Compute  $\mathcal{L}_{\text{geo}}$  on  $\mathcal{U}_t$  via Eqn. (17);
18     Update centroids  $\mathbf{M}_{t+1}$  via Eqn. (19);
19   Compute candidate parameters:
20      $\theta_t^+ = \theta_t - \eta \nabla_{\theta_t}(\mathcal{L}_{\text{trust}} + \mathcal{L}_{\text{geo}})$ ;
    // Stage 3: Retain source-compatible layer updates
21   for each layer  $l \in \{1, \dots, L\}$  do
22     Estimate curvature proxies  $I_t^l$  and  $I_0^l$  on current batch;
23     Compute retention gate  $\mu_t^l$  via Eqn. (23);
    Update layer  $\theta_{t+1}^l$  via Eqn. (21);

```

Output: Adapted model parameters θ_T

of test-time adaptation under both synthetic shifts and natural shifts.

Baselines. We compare DCF against representative source-free TTA baselines: BN Adapt [36], Tent [9], CoTTA [17], RoTTA [18], TRIBE [20], SPA [15], PTTA [16], SAR [14], DeYO [40], and AEA [19], along with the frozen source model (No Adapt). Specifically, BN Adapt [36] represents normalization-statistics adaptation; Tent [9] instantiates entropy-minimization TTA; to address continual or dynamic test streams, CoTTA [17] relies on mean-teacher/self-training, RoTTA [18] employs a memory bank, and TRIBE [20] adopts robust normalization; SPA [15] and PTTA [16] further study versatile/practical self-bootstrapping adaptation; SAR [14] improves robustness via sharpness-aware recovery, DeYO [40] utilizes disentangled-factor filtering, and AEA [19] leverages energy alignment. Unless otherwise specified, we use official implementations and hyperparameters for all baselines.

Models and Implementation Details. Following prior TTA works [17, 18, 20], we use a pre-trained ResNet-50 [1] backbone from RobustBench [48] and Torchvision [49], and set the batch size to 64 for all experiments. Unless otherwise specified, we follow the official implementation of Tent [9] and optimize the model with SGD (learning rate 2.5×10^{-4} , momentum 0.9). For hyperparameters in DCF, $(v_{PCS}, v_{Ent}, \mu_0, \tau)$ are set to $(0.15, 0.6, 0.99, 1.5)$. The OT objective in Eqn. (16) is solved via Sinkhorn iterations, and all results are averaged over 5 runs. All experiments are conducted on a single NVIDIA A100 GPU.

Scenarios. To rigorously evaluate robustness and adaptability under temporally correlated non-stationary image streams, we consider three synthetic test scenarios that simulate evolving acquisition and corruption conditions, illustrated in Figure 4:

- **Temporally-correlated Covariate Shifts (T-CS):** Covariate shifts occur sequentially across different corruption types, $\mathcal{P}_1^{C_1} \rightarrow \mathcal{P}_2^{C_2} \rightarrow \dots \rightarrow \mathcal{P}_t^{C_t}$, where each $\mathcal{P}_i^{C_i}$ represents the distribution under corruption type C_i with a fixed severity level (e.g., transitioning from motion blur to snow to fog, mimicking natural temporal variations).
- **Temporally-correlated Covariate and Label Shifts (T-CS-LS):** Covariate shift is combined with imbalanced label distribution shifts, $(\mathcal{P}_1^{C_1}, \pi_1^Y) \rightsquigarrow (\mathcal{P}_2^{C_2}, \pi_2^Y) \rightsquigarrow \dots \rightsquigarrow (\mathcal{P}_t^{C_t}, \pi_t^Y)$, where π_i^Y denotes the label distribution at time i sampled from a Dirichlet distribution (e.g., certain object classes becoming more/less frequent as environmental conditions change). We use the Dirichlet distribution with parameter $\delta = 0.1$ following [18] to simulate label shifts.
- **Temporally-correlated Covariate Shifts with Mixed Severity Levels (T-CS-MSL):** Covariate shifts with dynamically varying corruption severities, $\mathcal{P}_1^{C_1, \sigma_1} \rightarrow \mathcal{P}_2^{C_2, \sigma_2} \rightarrow \dots \rightarrow \mathcal{P}_t^{C_t, \sigma_t}$, where severity level σ_t is chosen uniformly at random from $\{1, \dots, 5\}$, creating a challenging stream where the model must handle both mild and severe shifts across the same temporal sequence.

Here, $\rightarrow$ denotes covariate shift and $\rightsquigarrow$ denotes covariate+label shift. To evaluate long-horizon stability, we include a **Long-horizon Temporally-correlated** test stream extended from the continually evolving setting in [17]. We formulate this extended sequence as $\mathcal{S}_{\text{long}} = \bigoplus_{r=1}^{15} (\mathcal{P}_{r,1} \rightarrow \dots \rightarrow \mathcal{P}_{r,15})$, exposing models to 1,125,000 samples sequentially across 225 dynamic corruptions on ImageNet-C. We set the duration rounds to 15, which is sufficient to reveal the tendency of baseline methods to collapse over time.

B. Main Results

Robustness on Temporally-Correlated Shifts. To evaluate the adaptability of DCF against continuous distribution shifts, we compare it with state-of-the-art TTA baselines across three temporally-correlated synthetic settings (T-CS, T-CS-LS, and T-CS-MSL) as well as two natural shift benchmarks (ImageNet-R and ImageNet-Sketch). As shown in Table I and Table III, DCF achieves the best performance among the compared methods across all settings.

- *Synthetic Corruptions (Table I):* Under the standard temporally-correlated covariate shift (T-CS), DCF achieves

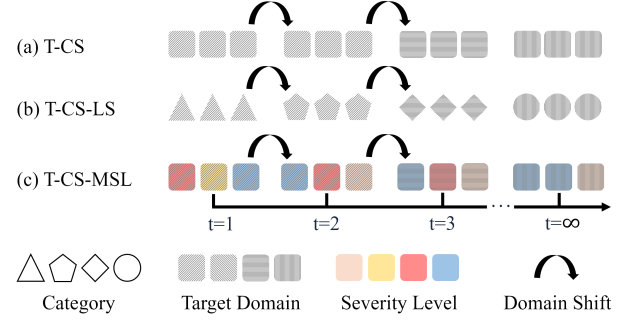


Fig. 4. Illustration of the challenging test data streams considered in our experiments. **T-CS** denotes temporally-correlated covariate shift, ***LS** denotes label shift, and ***MSL** denotes mixed severity levels.

the highest average accuracy of 43.96%, outperforming strong baselines like DeYO (42.35%) and SAR (40.25%). The performance gap widens on challenging structural shifts like IN-C-Bar, where our method yields 45.71% (vs. DeYO’s 44.78%). Crucially, when introducing complex joint shifts—such as Dirichlet label shifts (T-CS-LS) and dynamically mixed severities (T-CS-MSL)—DCF maintains robust superiority, delivering average accuracies of 43.84% and 55.27%, respectively.

- *Natural Shifts (Table III):* Beyond synthetic corruptions, DCF demonstrates strong generalization to real-world domain shifts. It reaches an average accuracy of 41.58% on IN-R and IN-Sketch, surpassing the second-best method (DeYO, 39.00%) by a significant margin of +2.58%.

These results suggest that decoupling sample routing, subset-specific adaptation, and layer-wise stabilization effectively mitigates the error accumulation typically induced by non-stationary data streams.

Stability over Long-Horizon Temporally Correlated Streams. A critical flaw of many TTA methods is their tendency to collapse when exposed to prolonged, non-stationary streams. To rigorously test this long-horizon stability, we conduct 15 continuous rounds of adaptation on ImageNet-C, exposing models to over 1.12 million samples sequentially.

As explicitly revealed by the Δ metric in Table II, unconstrained self-training updates (e.g., Tent, RoTTA, SPA) lead to catastrophic error accumulation. For instance, Tent plummets by 43.65% from Round 1 to Round 15 under T-CS-MSL. Surprisingly, even recent methods equipped with advanced sample filtering mechanisms (e.g., DeYO, PTTA) fail to prevent this late-stage collapse, turning from early gains to near-zero accuracy under T-CS. While conservative methods like CoTTA and SAR avoid severe drops, their absolute accuracy remains substantially lower.

In contrast, DCF consistently resists degradation across all long-horizon scenarios while maintaining the highest absolute accuracy at every single round. Rather than collapsing, our method’s performance trajectory remains resilient over the entire stream. This long-horizon evidence validates our core proposition: stable adaptation in non-stationary streams requires the joint, decoupled control of input-side evidence (routing) and layer-wise parameter drift (stabilization).

TABLE I

COMPARISON WITH STATE-OF-THE-ART METHODS W.R.T. AVERAGE ACCURACY (%) ACROSS ALL DOMAINS WITHIN EACH DATASET UNDER T-CS, T-CS-LS, AND T-CS-MSL SCENARIOS (IN-C, IN-3DCC, AND IN-C-BAR). BEST VALUES ARE SHOWN IN **BOLDFACE**.

Methods	T-CS				T-CS-LS				T-CS-MSL			
	IN-C	IN-3DCC	IN-C-Bar	Avg	IN-C	IN-3DCC	IN-C-Bar	Avg	IN-C	IN-3DCC	IN-C-Bar	Avg
No Adapt	7.19	14.35	10.75	10.76	7.19	14.37	10.75	10.77	18.27	24.75	20.83	21.28
BN Adapt ICML 15	29.83	34.61	39.02	34.49	29.66	34.55	39.12	34.44	45.45	49.30	50.71	48.49
Tent ICLR 21	37.55	39.47	43.03	40.02	37.30	39.17	42.89	39.79	50.76	52.48	53.75	52.33
CoTTA CVPR 22	31.77	36.08	40.13	35.99	31.58	36.06	40.16	35.93	47.27	50.41	51.65	49.78
RoTTA CVPR 23	31.28	36.24	37.66	35.06	31.55	36.01	37.83	35.13	47.38	50.27	51.12	49.59
SAR ICLR 23	38.26	39.47	43.01	40.25	38.17	39.44	42.92	40.18	50.90	52.62	53.75	52.42
DeYO ICLR 24	40.97	41.31	44.78	42.35	40.71	41.03	44.49	42.08	53.00	54.03	55.46	54.16
AEA ICLR 25	22.29	34.36	41.77	32.81	24.19	35.68	42.52	34.13	50.64	50.95	53.72	51.77
TRIBE AAAI 24	36.30	39.88	36.62	37.60	36.50	40.17	40.72	39.13	49.87	53.06	51.96	51.63
LAW WACV 24	33.30	37.12	41.46	37.29	35.84	38.49	42.76	39.03	50.22	52.46	53.49	52.06
SPA ICML 25	36.57	37.35	41.38	38.43	36.57	37.39	40.90	38.29	49.22	50.77	51.90	50.63
PTTA ICML 25	38.33	38.68	43.56	40.19	38.14	38.89	43.73	40.25	50.13	51.56	53.52	51.74
Ours	43.49	42.67	45.71	43.96	43.29	42.77	45.45	43.84	54.47	55.23	56.11	55.27

TABLE II

COMPARISON WITH STATE-OF-THE-ART METHODS ON IN-C UNDER LONG-HORIZON T-CS, T-CS-LS, AND T-CS-MSL. WE REPORT ROUND 1/7/15 RESULTS, THE MEAN OVER THESE THREE REPORTED ROUNDS (AVG.), AND Δ . Δ = ROUND-15 - ROUND-1 VALUES ARE COLORED **RED** IF NEGATIVE. BEST VALUES ARE SHOWN IN **BOLDFACE**.

Method	(a) Long-horizon T-CS					(b) Long-horizon T-CS-LS					(c) Long-horizon T-CS-MSL				
	Time →					Time →					Time →				
	Round 1	Round-7	Round-15	Avg.	Δ	Round 1	Round-7	Round-15	Avg.	Δ	Round 1	Round-7	Round-15	Avg.	Δ
No Adapt	7.19	7.19	7.19	7.19	0.00	7.19	7.19	7.19	7.19	0.00	18.27	18.02	18.28	18.19	0.01
BN Adapt ICML 15	29.83	29.83	29.83	29.83	0.00	29.66	29.66	29.66	29.66	0.00	45.45	45.39	45.66	45.50	0.21
Tent ICLR 21	37.55	9.32	0.49	15.79	-37.06	37.30	26.04	12.64	25.33	-24.66	50.76	43.99	7.11	33.95	-43.65
CoTTA CVPR 22	31.77	37.43	34.43	34.54	2.66	31.58	36.16	36.28	34.67	4.70	47.27	51.27	50.00	49.51	2.73
RoTTA CVPR 23	31.28	16.53	1.35	16.39	-29.93	31.55	30.83	16.56	26.31	-14.99	47.38	15.35	0.27	21.00	-47.11
SAR ICLR 23	38.26	39.99	38.45	38.90	0.19	38.17	40.13	39.31	39.20	1.14	50.90	53.63	53.80	52.78	2.90
DeYO ICLR 24	40.97	31.46	0.12	24.18	-40.85	40.71	33.43	15.84	29.99	-24.87	53.00	53.12	53.10	53.07	0.10
AEA ICLR 25	22.29	0.23	0.27	7.60	-22.02	24.19	3.82	1.87	9.96	-22.32	50.64	0.18	0.13	16.98	-50.51
TRIBE AAAI 24	36.30	19.40	19.51	25.07	-16.79	36.50	19.98	18.95	25.14	-17.55	49.87	42.62	42.27	44.92	-7.60
LAW WACV 24	33.30	32.46	13.87	26.54	-19.43	35.84	39.64	38.06	37.85	2.22	50.22	51.82	49.44	50.49	-0.78
SPA ICML 25	36.57	0.50	0.42	12.50	-36.15	36.57	10.31	4.97	17.28	-31.60	49.22	0.23	0.22	16.56	-49.00
PTTA ICML 25	38.33	29.06	0.51	22.63	-37.82	38.14	31.96	15.14	28.41	-23.00	50.13	50.40	50.55	50.36	0.42
Ours	43.49	43.11	43.84	43.48	0.35	43.29	43.14	43.37	43.27	0.08	54.47	54.47	54.65	54.53	0.18

TABLE III

COMPARISON WITH STATE-OF-THE-ART METHODS W.R.T. AVERAGE ACCURACY (%) UNDER NATURAL SHIFTS (IN-R AND IN-SKETCH). BEST VALUES ARE SHOWN IN **BOLDFACE**.

Methods	Natural Shift		
	IN-R	IN-Sketch	Avg.
No Adapt	36.39	14.61	25.50
BN Adapt ICML 15	39.66	25.29	32.48
Tent ICLR 21	42.82	30.51	36.67
CoTTA CVPR 22	40.25	26.30	33.28
RoTTA CVPR 23	39.63	28.12	33.88
SAR ICLR 23	42.99	31.89	37.44
DeYO ICLR 24	45.22	32.78	39.00
AEA ICLR 25	44.58	20.58	32.58
TRIBE AAAI 24	39.10	28.01	33.56
LAW WACV 24	42.44	29.92	36.18
SPA ICML 25	41.01	16.09	28.55
PTTA ICML 25	42.09	30.78	36.44
Ours	47.60	35.56	41.58

C. Ablations and Further Analysis

Effect of Components. To isolate the contribution of each module in DCF, we conduct a comprehensive ablation study as shown in Table IV. The unconstrained self-training baseline suffers from catastrophic error accumulation, dropping to 0.49% in the Long-horizon T-CS scenario (Avg. 31.53%). Introducing the trusted-evidence action (PSR) filters shortcut-driven predictions, improving the overall average to 37.25%, while the geometry-repair action (RGR) further boosts performance (Avg. 40.55%) by reusing routed-away samples to mitigate representation bias. However, sample-side control alone still degrades over extended streams, reaching only 26.25% at Round 15. Crucially, CLR restricts harmful parameter drift, preventing late-stage collapse (maintaining 43.84%) and achieving the highest overall average of 46.27%. Leave-one-out results show drops without PSR (41.43%) or RGR (45.36%), confirming that sample-level evidence routing and layer-level parameter stabilization are complementary and indispensable for robust adaptation.

TABLE IV
EFFECT OF COMPONENTS IN DCF ON IMAGENET-C. HERE, PSR, RGR, AND CLR DENOTE THE EQN. (10), EQN. (17), AND EQN. (21), RESPECTIVELY. WE ADDITIONALLY REPORT THE FINAL ACCURACY (ROUND 15) UNDER THE LONG-HORIZON T-CS SCENARIO TO EVALUATE LONG-HORIZON STABILITY.

PSR	RGR	CLR	T-CS	T-CS-LS	T-CS-MSL	Long-horizon T-CS	Avg
✗	✗	✗	37.55	37.30	50.76	0.49	31.53
✓	✗	✗	39.43	40.20	52.40	16.96	37.25
✓	✓	✗	41.37	41.13	53.45	26.25	40.55
✗	✓	✓	40.27	40.96	52.44	32.03	41.43
✓	✗	✓	41.87	42.71	53.97	42.88	45.36
✓	✓	✓	43.49	43.29	54.47	43.84	46.27

Effect of Design Choices in Stress Probe and Router. As shown in Figure 5, we further ablate the stress probe and routing rule in DCF. For the probe design, the Fourier-based operator achieves the best accuracy of 43.49%, outperforming pixel shuffle, low-pass filtering, and patch shuffle. This suggests that effective stress testing should not simply destroy image content, but should impose structured perturbations that preserve semantic layout while exposing shortcut-sensitive predictions. For the router design, our coupled entropy+PCS rule also obtains the highest accuracy of 43.49%, surpassing PCS-only routing as well as ETAGE-, SAR-, and DeYO-style variants [14, 21, 40]. The results indicate that PCS alone cannot exclude uncertain samples, while entropy- or gradient-based criteria alone may still retain confident but stress-inert predictions. The qualitative comparisons in Figure 6 further support this conclusion: the Fourier probe better preserves global object structure, and DCF produces more semantically focused response maps than Tent. Overall, these results validate that the Fourier stress probe and coupled entropy+PCS router provide safer adaptation evidence for stable test-time adaptation.

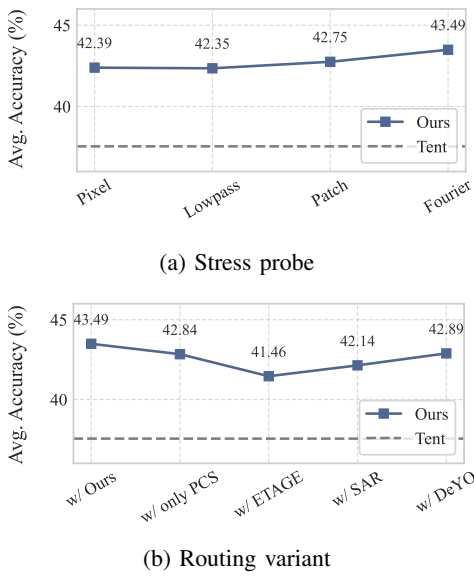


Fig. 5. **Ablation of design choices in stress probe and router.** Fourier probing and entropy-PCS routing perform best, showing that structured frequency cues and joint criteria provide more reliable adaptation signals.

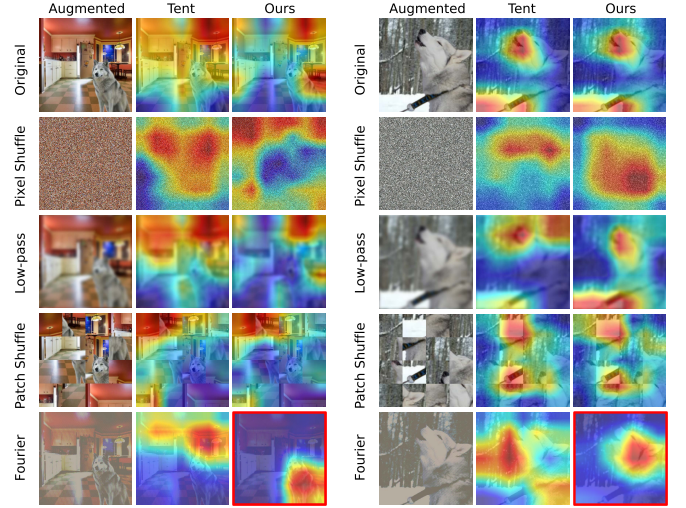


Fig. 6. **Qualitative comparison of stress probes and routing variants.** The left panel compares candidate stress probes, and the right panel compares routing variants. Both support the design choices used in DCF.

Pseudo-Label Purity and PCS-Entropy Space. To explain how our coupled routing mitigates confirmation bias, we evaluate pseudo-label purity on selected reliable samples. As shown in Table V, unconstrained self-training (Tent) attains only 42.7% purity, while entropy-only and PCS-only filtering provide modest improvements (44.0% and 47.1%). Combining both signals increases purity to 49.9%, indicating that PCS effectively removes confident but incorrect shortcut predictions missed by entropy. The PCS-Entropy distribution in Figure 7 further supports this finding: for DCF, reliable samples (low entropy, high PCS) are cleaner and better separated from spurious predictions across adaptation stages, whereas low-entropy, low-PCS regions contain many confident errors. In contrast, Tent lacks such structural constraints, causing representation degradation and weaker clean/noisy separation. These results validate our motivation (O1): confidence alone is insufficient for reliability, and the Fourier stress probe provides a critical geometric filter for stable adaptation.

TABLE V
PSEUDO-LABEL PURITY (%) OF SELECTED RELIABLE SAMPLES ON IMAGENET-C (T-CS). THE COUPLED ENTROPY+PCS ROUTER ACHIEVES THE HIGHEST PURITY, CONFIRMING THAT PCS EFFECTIVELY FILTERS OUT CONFIDENT YET INCORRECT PREDICTIONS THAT ENTROPY ALONE CANNOT EXCLUDE.

Method	Pseudo-label Purity (%)
Tent Baseline	42.7
Tent + Entropy Filter	44.0
Tent + PCS Filter	47.1
Tent + Entropy + PCS Filter	49.9

Layer-wise drift trajectories. Figure 8 provides complementary evidence for the parameter-side control view by tracking the layer-wise drift proxy $\varepsilon_{t,\text{block}}^{\text{par}} \propto \frac{1}{2} \|\theta_{t,\text{block}} - \theta_{0,\text{block}}\|_{l_0,\text{block}}^2$ over time. DCF exhibits a pulse-and-recover pattern: drift spikes at shift onsets are quickly suppressed by the stabilizer, whereas Tent shows cumulative growth, especially in deeper blocks.

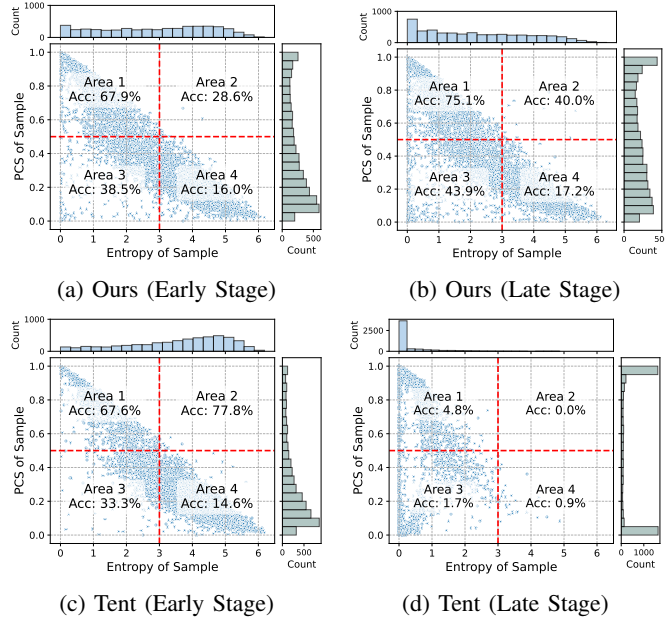


Fig. 7. **PCS-Entropy distribution of samples (ImageNet-C, long-horizon T-CS).** Our method (DCF) mainly maintains high accuracy in the trusted region (Area 1: low entropy, high PCS) across stages, effectively isolating confident but incorrect predictions into Area 3. In contrast, unconstrained self-training (Tent) suffers catastrophic collapse by the late stage.

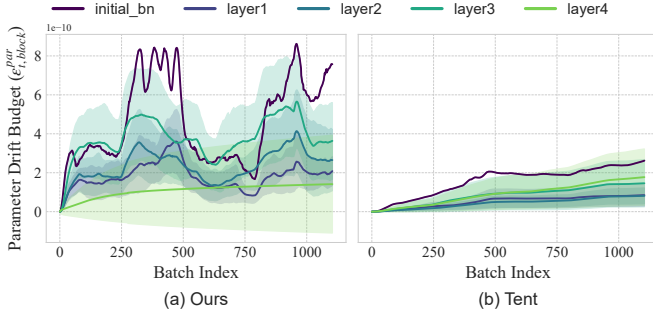


Fig. 8. **Layer-wise parameter drift on ImageNet-C (T-CS).** DCF shows a pulse-and-recover pattern, avoiding the progressive forgetting (drift accumulation) seen in Tent [9].

Static vs. Temporally-Correlated Streams. To highlight the specific challenge of temporal correlation, we compare adaptation performance under independently sampled (Static) and temporally-correlated (Long-horizon T-CS) test streams on ImageNet-C. As illustrated in Figure 9, many baselines that perform reasonably well under static conditions (e.g., Tent, DeYO, and AEA) suffer severe performance degradation or catastrophic collapse under T-CS, exposing their vulnerability to error accumulation. In contrast, our DCF not only achieves the highest accuracy in both settings but also maintains robust stability when transitioning to temporally correlated shifts. This confirms that our sample- and layer-aware control effectively prevents the compounding errors induced by non-stationary continuous streams.

Cross-Domain Transfer. Since TTA relies on self-training, it may over-specialize to narrow target distributions and thus degrade generalization to unseen domains. To assess cross-

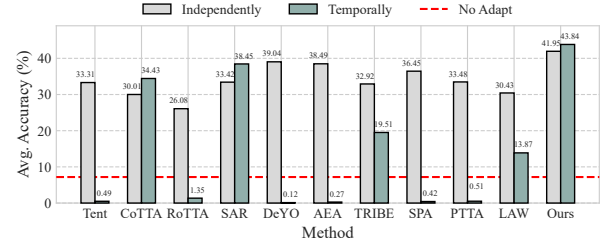


Fig. 9. **Static vs. Temporally-Correlated Streams.** Accuracy (%) on ImageNet-C under independently sampled test streams (Static, average accuracy) and temporally-correlated long-horizon test streams (Long-horizon T-CS, final round accuracy).

domain transfer, we adapt each method on one ImageNet-C corruption and evaluate it on the remaining 14 corruptions. As shown in Figure 10, DCF consistently achieves the largest average accuracy gains across all source domains, indicating that its control mechanisms help preserve more generalizable representations. Notably, methods like RoTTA show large gains when adapted to certain source domains but suffer significant drops when evaluated on others, suggesting overfitting to specific shifts. In contrast, DCF maintains positive gains across all source domains, demonstrating its ability to adapt stably while retaining transferable features that generalize well across diverse unseen shifts.

Tent	27.45	26.95	27.04	27.06	27.10	26.72	25.73	25.69	28.14	24.31	21.83	27.49	24.56	24.81	26.96	26.12
CoTTA	25.10	24.85	24.82	25.79	25.39	24.79	24.73	24.16	26.86	22.99	22.76	25.69	23.68	24.28	26.11	24.80
RoTTA	-3.14	-3.37	-4.94	-1.95	-1.85	0.47	2.94	1.58	2.56	-2.30	-0.24	-5.21	4.11	6.19	5.58	0.03
SAR	26.22	26.57	26.47	25.52	25.72	26.35	25.10	24.87	27.26	23.69	21.41	27.81	23.99	24.14	26.42	25.44
DeYO	28.65	27.80	28.69	28.40	28.77	29.68	27.34	26.64	30.33	25.10	23.10	27.80	26.43	27.37	29.07	27.68
TRIBE	27.48	26.73	26.43	26.97	26.16	26.23	27.02	25.60	25.85	24.08	25.18	26.42	26.15	26.20	27.61	26.27
AEA	29.09	28.34	28.90	29.07	28.71	29.41	27.79	27.33	30.58	25.29	23.92	26.84	26.75	27.70	29.61	27.95
SPA	27.18	28.14	27.88	28.72	28.32	26.60	28.43	26.94	28.94	28.19	24.34	24.39	27.40	28.68	30.40	27.64
PTTA	26.21	27.11	27.10	27.21	25.83	26.12	27.16	26.17	28.25	26.77	25.04	23.07	25.86	26.31	27.33	26.37
Ours	31.12	30.40	30.41	29.65	30.51	31.58	30.31	29.56	33.39	27.95	28.11	31.17	31.04	30.87	28.37	30.30
	Gaussian	Motion	Snow	Fog	Shot	Defocus	Contrast	Zoom	Brightness	Frost	Elastic	Glass	Pixdate	JPEG	Impulse	Avg
	Adaptation Domain															

Fig. 10. **Cross-domain Generalization.** Each cell shows the average accuracy improvement compared to No Adapt when a method (row) is adapted to a source domain (column) and then evaluated on the remaining 14 unseen ImageNet-C domains.

Generalizability across Architectures. To evaluate the generalizability of DCF across different model architectures, we test it on a range of backbones including ResNet-18 (RN-18), WideResNet-50 (WRN-50), ResNeXt-50 (RNx-50), ResNet-101 (RN-101), and ViT-B/16. As shown in Table VI, DCF consistently outperforms all baselines across these architectures under the T-CS scenario on ImageNet-C. Notably, the performance gap is particularly pronounced for larger models like ResNet-101 and ViT-B/16, where many baselines struggle to maintain stability. This suggests that our method’s control mechanisms effectively mitigate error accumulation even as model capacity increases, supporting its broad applicability in real-world deployment settings with diverse architectures.

TABLE VI

ARCHITECTURAL GENERALIZABILITY. DCF CONSISTENTLY SURPASSES BASELINES ON IMAGENET-C UNDER T-CS ACROSS DIVERSE BACKBONES. '/' INDICATES ARCHITECTURAL INCOMPATIBILITY.

Methods	RN-18	WRN-50	RNX-50	RN-101	ViT-B/16	Avg.
No Adapt	5.55	21.41	21.51	9.87	40.49	19.77
BN Adapt	25.78	33.86	32.94	32.03	40.49	33.02
Tent	31.13	43.45	42.41	41.22	45.43	40.73
CoTTA	26.97	37.27	35.56	34.86	40.32	35.00
RoTTA	24.51	37.81	36.14	35.54	/	33.50
SAR	31.74	43.48	42.88	42.70	45.42	41.24
DeYO	32.54	45.63	45.43	43.01	48.38	43.00
AEA	19.20	31.47	27.03	20.16	36.26	26.82
TRIBE	29.81	40.23	39.55	24.21	/	33.45
LAW	27.61	38.28	40.37	36.73	25.27	33.65
SPA	24.87	27.62	31.16	41.22	49.81	34.94
PTTA	31.20	42.80	40.50	41.92	47.08	40.70
Ours	34.25	47.47	46.86	47.67	54.95	46.24

Hyperparameter Sensitivity. We analyze the joint landscapes of key control hyperparameters in Figure 11. For sample routing, moderate thresholds (v_{PCS} , v_{Ent}) best balance trusted-evidence purity and self-supervision volume. For parameter retention, performance is relatively stable to the weighting sharpness τ but sensitive to the fixed retention factor μ : small μ limits plasticity, whereas $\mu \rightarrow 1.0$ amplifies error accumulation. Notably, our dynamic layer-wise stabilizer (CLR) surpasses the best static- μ baseline, reaching a peak average accuracy of 43.49%. This confirms the benefit of depth-aware, heterogeneous parameter control and supports our motivation (O3). Furthermore, DCF achieves a strong accuracy-runtime trade-off (Figure 12): processing an image in 0.005s (A100 GPU), it rivals DeYO and SAR in speed while outpacing RoTTA, TRIBE, and SPA.

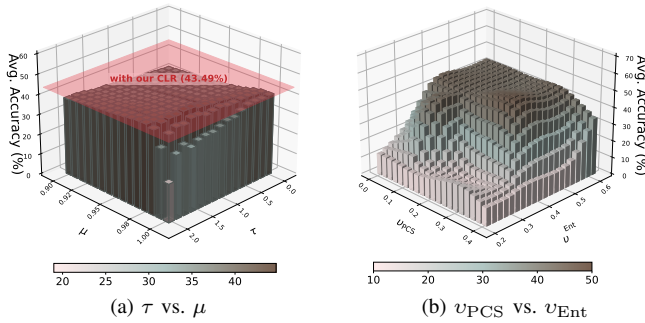


Fig. 11. **Hyperparameter sensitivity.** (a) PCS sharpness τ and fixed retention factor μ show a non-monotonic joint effect, highlighting the limitations of static retention and motivating the dynamic layer-wise retention in DCF (red plane). (b) Moderate PCS and entropy thresholds form a broad high-performing region, whereas overly loose or strict routing impairs adaptation.

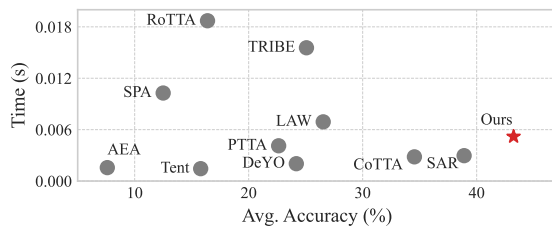


Fig. 12. **Accuracy-runtime trade-off.** Per-image GPU runtime vs. average accuracy on ImageNet-C (long-horizon T-CS). Better methods appear in the lower-right quadrant. Runtimes are averaged over 100 runs (single A100 GPU).

V. CONCLUSIONS

This paper reveals that the instability of test-time adaptation under temporally correlated shifts stems from a homogeneous treatment of intrinsically heterogeneous target evidence and layer-wise drift. To address this, we propose DCF, which decouples sample routing, subset-specific sample-side adaptation, and layer-aware parameter stabilization. By separating the routing decision from the adaptation objective, constraining confident but shortcut-driven updates, and repairing target geometry via optimal transport, DCF mitigates the confirmation bias that typically drives model collapse. Extensive evaluations show that DCF effectively halts the compounding errors that plague existing TTA methods in long-horizon deployment streams.

REFERENCES

- [1] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *CVPR*, 2016, pp. 770–778.
- [2] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, “An image is worth 16x16 words: Transformers for image recognition at scale,” in *ICLR*, 2021.
- [3] Y. Ganin and V. Lempitsky, “Unsupervised domain adaptation by backpropagation,” in *ICML*, 2015, pp. 1180–1189.
- [4] N. Courty, R. Flamary, D. Tuia, and A. Rakotomamonjy, “Optimal transport for domain adaptation,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 39, no. 9, pp. 1853–1865, 2017.
- [5] J. Liang, D. Hu, and J. Feng, “Do we really need to access the source data? Source hypothesis transfer for unsupervised domain adaptation,” in *ICML*, vol. 119, pp. 6028–6039, 18 Jul 2020.
- [6] J. Tian, J. Zhang, W. Li, and D. Xu, “Vdm-da: Virtual domain modeling for source data-free domain adaptation,” *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 32, no. 6, pp. 3749–3760, Jun. 2022.
- [7] Y. Zhao, Z. Zhong, Z. Luo, G. H. Lee, and N. Sebe, “Source-free open compound domain adaptation in semantic segmentation,” *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 32, no. 10, pp. 7019–7032, Oct. 2022.
- [8] Y. Xu, J. Yang, H. Cao, K. Wu, M. Wu, Z. Li, and Z. Chen, “Multi-source video domain adaptation with temporal attentive moment alignment network,” *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 33, no. 8, pp. 3860–3871, Aug. 2023.
- [9] D. Wang, E. Shelhamer, S. Liu, B. Olshausen, and T. Darrell, “Tent: Fully test-time adaptation by entropy minimization,” in *ICLR*, 2021.
- [10] Y. Sun, X. Wang, Z. Liu, J. Miller, A. Efros, and M. Hardt, “Test-time training with self-supervision for generalization under distribution shifts,” in *ICML*. PMLR, 2020, pp. 9229–9248.
- [11] J. Liu and Z. Yang, “Test-time adaptation for real-world video adverse weather restoration with meta batch normalization,” *IEEE Transactions on Circuits and Systems*

- for *Video Technology*, vol. 35, no. 6, pp. 5533–5544, Jun. 2025.
- [12] C. Zhang, F. Yang, C. Cui, S. Gong, W. Wang, X. Lin, Y. Qi, and L. Zhu, “Collaborative model and data adaptation at test time,” *IEEE Transactions on Circuits and Systems for Video Technology*, pp. 1–1, 2026.
 - [13] S. Niu, J. Wu, Y. Zhang, Y. Chen, S. Zheng, P. Zhao, and M. Tan, “Efficient test-time model adaptation without forgetting,” in *ICML*, vol. 162. PMLR, 17–23 Jul 2022, pp. 16 888–16 905.
 - [14] S. Niu, J. Wu, Y. Zhang, Z. Wen, Y. Chen, P. Zhao, and M. Tan, “Towards stable test-time adaptation in dynamic wild world,” in *ICLR*, 2023.
 - [15] S. Niu, G. Chen, P. Zhao, T. Wang, P. Wu, and Z. Shen, “Self-bootstrapping for versatile test-time adaptation,” in *ICML*, ser. Proceedings of Machine Learning Research, vol. 267. PMLR, 13–19 Jul 2025, pp. 46 611–46 628.
 - [16] J. Ma, H. Li, and X. Xiang, “PTTA: Purifying malicious samples for test-time model adaptation,” in *ICML*, ser. Proceedings of Machine Learning Research, vol. 267. PMLR, 13–19 Jul 2025, pp. 42 030–42 061.
 - [17] Q. Wang, O. Fink, L. Van Gool, and D. Dai, “Continual test-time domain adaptation,” in *CVPR*, 2022, pp. 7201–7211.
 - [18] L. Yuan, B. Xie, and S. Li, “Robust test-time adaptation in dynamic scenarios,” in *CVPR*, 2023, pp. 15 922–15 932.
 - [19] W. Choi, D.-Y. Kim, J. Park, J. Lee, Y. Park, D.-J. Han, and J. Moon, “Adaptive energy alignment for accelerating test-time adaptation,” in *ICLR*, 2025.
 - [20] Y. Su, X. Xu, and K. Jia, “Towards real-world test-time adaptation: Tri-net self-training with balanced normalization,” *AAAI*, vol. 38, no. 13, pp. 15 126–15 135, Mar. 2024.
 - [21] A. Shamsi, R. Becirovic, A. Argha, E. Abbasnejad, H. Alinejad-Rokny, and A. Mohammadi, “ETAGE: Enhanced test time adaptation with integrated entropy and gradient norms for robust model performance,” in *SMC*, 2025, pp. 5907–5911.
 - [22] J. Wang, C. Lan, C. Liu, Y. Ouyang, T. Qin, W. Lu, Y. Chen, W. Zeng, and P. S. Yu, “Generalizing to unseen domains: A survey on domain generalization,” *IEEE Transactions on Knowledge and Data Engineering*, vol. 35, no. 8, pp. 8052–8072, 2023.
 - [23] I. Ng, Y. Li, Z. Li, Y. Zheng, G. Chen, and K. Zhang, “A general representation-based approach to multi-source domain adaptation,” in *ICML*, 2025.
 - [24] D. Kim, K. Wang, S. Sclaroff, and K. Saenko, “A broad study of pre-training for domain generalization and adaptation,” in *ECCV*, 2022.
 - [25] S. J. Pan, I. W. Tsang, J. T. Kwok, and Q. Yang, “Domain adaptation via transfer component analysis,” *IEEE transactions on neural networks*, vol. 22, no. 2, pp. 199–210, 2010.
 - [26] B. Sun and K. Saenko, “Deep coral: Correlation alignment for deep domain adaptation,” in *ECCV*. Springer, 2016, pp. 443–450.
 - [27] H. Yan, Y. Ding, P. Li, Q. Wang, Y. Xu, and W. Zuo, “Mind the class weight bias: Weighted maximum mean discrepancy for unsupervised domain adaptation,” in *CVPR*, 2017, pp. 2272–2281.
 - [28] G. Kang, L. Jiang, Y. Wei, Y. Yang, and A. G. Hauptmann, “Contrastive adaptation network for single-and multi-source domain adaptation,” *IEEE transactions on pattern analysis and machine intelligence*, 2020.
 - [29] Y. Ganin, E. Ustinova, H. Ajakan, P. Germain, H. Larochelle, F. Laviolette, M. Marchand, and V. Lempitsky, “Domain-adversarial training of neural networks,” *The journal of machine learning research*, vol. 17, no. 1, pp. 1–35, 2016.
 - [30] L. Chen, H. Chen, Z. Wei, X. Jin, X. Tan, Y. Jin, and E. Chen, “Reusing the task-specific classifier as a discriminator: Discriminator-free adversarial domain adaptation,” in *ICCV*, 2022, pp. 7181–7190.
 - [31] Y. Zou, Z. Yu, X. Liu, B. Kumar, and J. Wang, “Confidence regularized self-training,” in *ICCV*, 2019, pp. 5982–5991.
 - [32] K. Mei, C. Zhu, J. Zou, and S. Zhang, “Instance adaptive self-training for unsupervised domain adaptation,” in *ECCV*. Springer, 2020, pp. 415–430.
 - [33] H. Liu, J. Wang, and M. Long, “Cycle self-training for domain adaptation,” *NeurIPS*, vol. 34, pp. 22 968–22 981, 2021.
 - [34] M. Litrico, A. Del Bue, and P. Morerio, “Guiding pseudo-labels with uncertainty estimation for source-free unsupervised domain adaptation,” in *CVPR*, 2023, pp. 7640–7650.
 - [35] T. Chu, Y. Liu, J. Deng, W. Li, and L. Duan, “Denoised maximum classifier discrepancy for sourcefree unsupervised domain adaptation,” in *AAAI*, vol. 2, 2022.
 - [36] S. Ioffe and C. Szegedy, “Batch normalization: Accelerating deep network training by reducing internal covariate shift,” in *ICML*. pmlr, 2015, pp. 448–456.
 - [37] T. Gong, J. Jeong, T. Kim, Y. Kim, J. Shin, and S.-J. Lee, “NOTE: Robust continual test-time adaptation against temporal correlation,” in *NeurIPS*, A. H. Oh, A. Agarwal, D. Belgrave, and K. Cho, Eds., 2022.
 - [38] O. Press, S. Schneider, M. Kümmerer, and M. Bethge, “Rdumb: A simple approach that questions our progress in continual test-time adaptation,” in *NeurIPS*, vol. 36, 2023, pp. 39 915–39 935.
 - [39] T. Gong, Y. Kim, T. Lee, S. Chottananurak, and S.-J. Lee, “Sotta: Robust test-time adaptation on noisy data streams,” *NeurIPS*, vol. 36, pp. 14 070–14 093, 2023.
 - [40] J. Lee, D. Jung, S. Lee, J. Park, J. Shin, U. Hwang, and S. Yoon, “Entropy is not enough for test-time adaptation: From the perspective of disentangled factors,” in *ICLR*, 2024.
 - [41] J. Park, J. Kim, H. Kwon, I. Yoon, and K. Sohn, “Layer-wise auto-weighting for non-stationary test-time adaptation,” in *WACV*, January 2024, pp. 1414–1423.
 - [42] Y. Lee, A. S. Chen, F. Tajwar, A. Kumar, H. Yao, P. Liang, and C. Finn, “Surgical fine-tuning improves adaptation to distribution shifts,” in *ICLR*, 2023.
 - [43] D. Hendrycks and T. Dietterich, “Benchmarking neural network robustness to common corruptions and perturbations,” in *ICLR*, 2019.
 - [44] O. F. Kar, T. Yeo, A. Atanov, and A. Zamir, “3D common corruptions and data augmentation,” in *CVPR*, 2022, pp.

18 963–18 974.

- [45] E. Mintun, A. Kirillov, and S. Xie, “On interaction between augmentations and corruptions in natural corruption robustness,” in *NeurIPS*, A. Beygelzimer, Y. Dauphin, P. Liang, and J. W. Vaughan, Eds., 2021.
- [46] D. Hendrycks, S. Basart, N. Mu, S. Kadavath, F. Wang, E. Dorundo, R. Desai, T. Zhu, S. Parajuli, M. Guo, D. Song, J. Steinhardt, and J. Gilmer, “The many faces of robustness: A critical analysis of out-of-distribution generalization,” in *ICCV*, 2021, pp. 8320–8329.
- [47] H. Wang, S. Ge, Z. Lipton, and E. P. Xing, “Learning robust global representations by penalizing local predictive power,” in *NeurIPS*, 2019, pp. 10 506–10 518.
- [48] F. Croce, M. Andriushchenko, V. Sehwag, E. Debenedetti, N. Flammarion, M. Chiang, P. Mittal, and M. Hein, “RobustBench: A standardized adversarial robustness benchmark,” in *NeurIPS Datasets and Benchmarks Track*, 2021.
- [49] TorchVision Team, “TorchVision: PyTorch’s computer vision library,” GitHub repository, 2016.



Haoyu Xiong received the M.S. degree in electronic information from Yunnan Normal University, Kunming, China, in 2024. He is currently working towards the Ph.D. degree in the School of Information Science and Engineering of Yunnan University, Kunming, China. His research interests include test-time domain adaptation, domain generalization, and robust visual recognition. He has authored or coauthored papers in leading conferences, including the ICML and so on.



Chengchao Wang received the B.S. degree in School of Automation and Electrical Engineering from Shenyang Ligong University, Shenyang, China, in 2019, and the master’s degree from the School of Information Science and Engineering of Yunnan University, Kunming, China in 2022. He is currently working towards the Ph.D. degree in the School of Information Science and Engineering of Yunnan University, Kunming, China. His current research interests include image processing and deep learning.



Yang Li is currently an Assistant Researcher with Yunnan University, School of Information Science and Engineering, and is pursuing the Ph.D. degree in information and communication engineering at Yunnan University, Kunming, China. He received the M.S. degree in electronic and communication engineering from Yunnan University in 2012 and the B.Eng. degree in electronic information engineering from Changchun Institute of Technology in 2010.



Qiuxia Yang received the Ph.D. degree from Yunnan University, Kunming, China. She is currently a Lecturer with the School of Engineering, Yunnan University. Her research interests include image generation and computer vision.



Wenhan Wu received the B.Eng. degree from the University of Electronic Science and Technology of China in 2018, the M.S. degree from Stevens Institute of Technology in 2020, and the Ph.D. degree from the University of North Carolina at Charlotte in 2025. He is currently a Lecturer with the Yunnan-Malaya Institute, Yunnan University, China. His research interests include deep learning, computer vision, video understanding, action recognition, motion analysis, and pose estimation.



Zhengpeng Zhao was born in Yunnan, China, in 1972. He received the B.Sc. and M.Sc. degrees in electronic engineering from Yunnan University, Kunming, China, in 1993 and 2004, respectively. His research interests include image compression and signal processing.



Yuanyuan Pu was born in Kunming, Yunnan, China, in 1972. She received the B.S., M.S., and Ph.D. degrees from Yunnan University, Kunming, in 1995, 1998, and 2011, respectively. She is currently a Professor with the School of Engineering, Yunnan University, and the Key Lab. of IoT Technology and Application in Yunnan Province, Kunming. She has authored or coauthored over 30 scientific articles, including the TMM, TNNLS, and so on. Her main research interests include digital image processing and computer vision.



Jinde Cao (Fellow, IEEE) received the B.S. degree from Anhui Normal University, Wuhu, China, the M.S. degree from Yunnan University, Kunming, China, and the Ph.D. degree from Sichuan University, Chengdu, China, all in mathematics/applied mathematics, in 1986, 1989, and 1998, respectively. He is an Endowed Chair Professor, the Dean of the School of Mathematics, the Director of the Jiangsu Provincial Key Laboratory of Networked Collective Intelligence of China and the Director of the Research Center for Complex Systems and Network Sciences at Southeast University. Prof. Cao was a recipient of the National Innovation Award of China, Gold medal of Russian Academy of Natural Sciences, Obada Prize and the Highly Cited Researcher Award in Engineering, Computer Science, and Mathematics by Thomson Reuters/Clarivate Analytics. He is elected as a member of the Academy of Europe, a foreign member of Russian Academy of Engineering, a member of the European Academy of Sciences and Arts, a foreign member of Russian Academy of Natural Sciences, a foreign fellow of Pakistan Academy of Sciences, a fellow of African Academy of Sciences, a foreign Member of the Lithuanian Academy of Sciences, an IASCYS academician, and a full member of Sigma Xi.